Light

GCSE Computer Science — CPU Architecture (Set 3)

Over **1,100,000+ marks** of exam-aligned content across **67 qualifications**. 4,200+ auto-marked assessments & long-answer practice. No login required.

GCSE • Edexcel • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

00:00 Start Pause Reset

← Back to Cpu Architecture

Multiple-choice topic test for **GCSE Computer Science (Edexcel)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

A program calculates a student's total score by adding marks from three tests.
Which component of the CPU performs this addition?

- ○ **A.** The Control Unit (CU)
- ○ **B.** The Program Counter (PC)
- ○ **C.** The Memory Data Register (MDR)
- ○ **D.** The Arithmetic Logic Unit (ALU)

Q2 (1 mark)

A machine code instruction contains two parts: one part tells the CPU what operation to perform, and another part specifies the data or address to use.

What is the part that tells the CPU what operation to perform called?

- ○ **A.** The opcode
- ○ **B.** The operand
- ○ **C.** The register
- ○ **D.** The address

Q3 (1 mark)

A textbook describes the 'system bus' as connecting the CPU to memory and other components.
Which statement correctly describes what the system bus consists of?

- ○ **A.** The system bus is another name for the data bus only.
- ○ **B.** The system bus is a wireless connection between components.
- ○ **C.** The system bus is a backup bus used only when other buses fail.
- ○ **D.** The system bus consists of the data bus, address bus, and control bus together.

Q4 (1 mark)

The CPU continuously processes instructions using a cycle with three main stages.
What is the correct order of these stages?

- ○ **A.** Decode, Fetch, Execute
- ○ **B.** Execute, Decode, Fetch
- ○ **C.** Fetch, Execute, Decode

- ○ **D.** Fetch, Decode, Execute

Q5 (1 mark)

In Von Neumann architecture, the CPU fetches instructions from a component called main memory.
What is another name for main memory?

- ○ **A.** RAM (Random Access Memory)
- ○ **B.** Hard disk drive
- ○ **C.** USB flash drive
- ○ **D.** CD-ROM

Q6 (1 mark)

Von Neumann architecture stores both programs and data in the same memory.
What is one limitation of this design?

- ○ **A.** The CPU cannot fetch an instruction and data at the same time using the same bus
- ○ **B.** Programs cannot be larger than 1 MB in size
- ○ **C.** Only one program can ever be stored in memory
- ○ **D.** Data must be deleted before a new program can run

Q7 (1 mark)

Inside a CPU, a component generates a regular signal that looks like a square wave, switching between high and low voltage at a constant rate.

What is the purpose of this signal?

- ○ **A.** To provide power to the CPU transistors
- ○ **B.** To encrypt data as it moves through the CPU
- ○ **C.** To synchronise the timing of all CPU operations
- ○ **D.** To cool the CPU by creating airflow

Q8 (1 mark)

A student unplugs their computer to move it to another room. When they plug it back in and turn it on, the data that was in the CPU registers is gone.
What property of registers does this demonstrate?

- ○ **A.** Registers are non-volatile and should have retained the data
- ○ **B.** Registers are read-only and cannot store user data
- ○ **C.** Registers are volatile and lose their contents when power is removed
- ○ **D.** Registers only work when connected to the internet

Q9 (1 mark)

After decoding an instruction, the CPU needs to coordinate which components should be active and what signals should be sent.
Which component generates these control signals?

- ○ **A.** The Control Unit (CU)
- ○ **B.** The Program Counter (PC)
- ○ **C.** The Accumulator (ACC)

- ○ **D.** The Memory Address Register (MAR)

Q10 (1 mark)

The Control Unit (CU) is one of the main components inside the CPU. What is the main role of the Control Unit?

- ○ **A.** To perform arithmetic calculations like addition and subtraction
- ○ **B.** To store the results of calculations
- ○ **C.** To coordinate and control all CPU operations by sending control signals
- ○ **D.** To permanently store programs and data

Q11 (1 mark)

During program execution, the CPU reads an instruction from RAM. The instruction travels from RAM and arrives at the CPU.
Which register first receives this instruction when it arrives at the CPU?

- ○ **A.** The Current Instruction Register (CIR)
- ○ **B.** The Program Counter (PC)
- ○ **C.** The Memory Data Register (MDR)
- ○ **D.** The Memory Address Register (MAR)

Q12 (1 mark)

The CPU contains a clock that generates regular electrical pulses.
What is the purpose of these clock pulses?

- ○ **A.** To synchronise all operations within the CPU
- ○ **B.** To display the current time on the computer screen
- ○ **C.** To count how many programs have been run
- ○ **D.** To provide power to the CPU transistors

Q13 (2 marks)

A machine code instruction uses 16 bits total: 4 bits for the opcode and 12 bits for the operand.
Which statement correctly describes what this instruction format allows?

- ○ **A.** The instruction can specify one of 16 different operations and reference memory addresses up to 4095.
- ○ **B.** The instruction can specify one of 4 different operations and reference memory addresses up to 12.
- ○ **C.** The instruction can specify one of 4096 different operations and reference memory addresses up to 15.
- ○ **D.** The instruction can only perform addition because 16 bits is too small for other operations.

Q14 (2 marks)

During normal program execution, the Program Counter (PC) is incremented after each instruction is fetched. However, some instructions cause the PC to change in a different way.
Which type of instruction would cause the PC to be loaded with a completely new address rather than just being incremented?

- ○ **A.** An ADD instruction that adds two numbers together
- ○ **B.** A COMPARE instruction that checks if two values are equal

○ **C.** A LOAD instruction that reads data from memory into a register

○ **D.** A JUMP instruction that transfers control to a different part of the program

Q15 (2 marks)

A CPU designer is choosing between a 20-bit address bus and a 24-bit address bus for a new processor.
Which statement correctly describes the effect of this choice?

○ **A.** The 24-bit bus can transfer data 4 bits faster per operation than the 20-bit bus.

○ **B.** Both buses allow exactly the same amount of memory to be addressed because address bus width does not affect memory capacity.

○ **C.** The 20-bit bus is better because it uses less electricity and generates less heat.

○ **D.** The 24-bit bus allows the CPU to address 16 times more memory locations than the 20-bit bus.

Q16 (2 marks)

A CPU has several specialised registers: PC, MAR, MDR, CIR, and ACC.
Which statement correctly pairs each register with its primary function?

○ **A.** PC stores calculation results; ACC holds the next instruction address; MAR and MDR both store control signals

○ **B.** CIR stores data from RAM; MDR holds the current instruction; PC and MAR both track program flow

○ **C.** PC holds the next instruction address; MAR holds the memory address being accessed; MDR holds data being transferred; CIR holds

the instruction being executed; ACC stores calculation results

○ **D.** All five registers have identical functions and can be used interchangeably by the Control Unit

Q17 (2 marks)

A CPU has a clock speed of 3.2 GHz.
What does this specification tell us about the CPU?

○ **A.** The CPU can store 3.2 gigabytes of data in its registers

○ **B.** The CPU can run 3.2 billion different programs simultaneously

○ **C.** The CPU has 3.2 billion transistors inside it

○ **D.** The CPU's clock generates 3.2 billion pulses per second to synchronise operations

Q18 (2 marks)

Registers are a type of memory located inside the CPU.
Which statement correctly compares registers to RAM?

○ **A.** Registers are slower than RAM but can store more data

○ **B.** Registers are used for permanent storage while RAM is temporary

○ **C.** Registers and RAM have identical speed and capacity

○ **D.** Registers are faster than RAM but can only store small amounts of data

Q19 (2 marks)

During the execute stage of an ADD instruction, the ALU needs two values to add together.
Where do these values typically come from?

- ○ **A.** Both values must come directly from RAM during the execute stage.
- ○ **B.** Values are entered by the user through the keyboard during execution.
- ○ **C.** The ALU generates both values internally based on the opcode.
- ○ **D.** Values come from registers or the MDR, having been loaded in previous instructions or as part of this instruction.

Q20 (2 marks)

A student states: 'The data bus is bidirectional, but the address bus is unidirectional.'
Which explanation correctly justifies why these buses have different directional properties?

- ○ **A.** This statement is incorrect; both buses are bidirectional because information must flow both ways.
- ○ **B.** Both buses are unidirectional; data only flows from CPU to memory.
- ○ **C.** The address bus is bidirectional and the data bus is unidirectional; the student has them reversed.
- ○ **D.** Data needs to flow both to and from memory (read and write operations), but addresses are always specified by the CPU, so they only need to flow one way.

Q21 (2 marks)

During the decode stage of the FDE cycle, the instruction in the CIR is interpreted.
What happens during this interpretation?

- ○ **A.** The binary instruction is converted to English text for the user to read
- ○ **B.** The Control Unit determines what operation the instruction requires and prepares control signals
- ○ **C.** The instruction is sent back to RAM for permanent storage
- ○ **D.** The ALU performs the calculation specified by the instruction

Q22 (2 marks)

A traffic light uses an embedded system to control when each light turns on and off.
Which characteristics make an embedded system suitable for this application?

- ○ **A.** The ability to run web browsers and office applications
- ○ **B.** Dedicated function, high reliability, and continuous operation without user intervention
- ○ **C.** Large storage capacity for video files and photographs
- ○ **D.** High-resolution graphics output for displaying advertisements

Q23 (2 marks)

A computer system has three main buses: address bus, data bus, and control bus.
Which description correctly matches each bus to its function?

- ○ **A.** Address bus carries data; data bus carries addresses; control

bus carries clock signals only

○ **B.** All three buses carry the same information but provide redundancy for reliability

○ **C.** Address bus specifies memory locations; data bus carries data values; control bus carries control signals like read/write

○ **D.** Address bus connects to the keyboard; data bus connects to the screen; control bus connects to the mouse

Q24 (3 marks)

A computer system has a 20-bit address bus.
Which calculation correctly determines the maximum amount of memory this system can address?

○ **A.** 20 bytes, because each address bit represents one byte

○ **B.** 1,048,576 bytes (1 MiB), because 2^20 = 1,048,576 addressable locations

○ **C.** 160 bytes, because 20 bits × 8 bits per byte = 160

○ **D.** 20 MB, because address bus width in bits equals memory in megabytes

Q25 (3 marks)

A CPU is executing a program. The PC currently contains address 1000. The instruction at address 1000 is a simple ADD instruction that does not involve memory access.
What will the PC contain after this instruction completes?

○ **A.** 1000, because the PC does not change during normal execution

- ○ **B.** 1001, because the PC is incremented to point to the next instruction
- ○ **C.** The result of the addition, because ADD instructions modify the PC
- ○ **D.** 0, because the PC resets after each instruction completes

Q26 (3 marks)

A CPU needs to transfer 512 bytes of data from RAM. The system has a 32-bit data bus.
How many separate bus transfers are required?

- ○ **A.** 16 transfers, because 512 ÷ 32 = 16
- ○ **B.** 128 transfers, because 512 bytes ÷ 4 bytes per transfer = 128
- ○ **C.** 4096 transfers, because 512 × 8 bits = 4096 bit transfers
- ○ **D.** 512 transfers, because each byte requires exactly one transfer

Q27 (3 marks)

A car manufacturer needs to choose a processor for the anti-lock braking system (ABS) in their vehicles.
Which factors should be prioritised when selecting a processor for this safety-critical embedded application?

- ○ **A.** High graphics performance and large storage for multimedia features
- ○ **B.** Extremely high reliability, fast response time, and ability to operate in harsh conditions (temperature, vibration)
- ○ **C.** Ability to connect to social media and run smartphone applications

○ **D.** Support for multiple monitors and high-resolution video output

Q28 (3 marks)

A computer system has a 16-bit address bus. The system designer needs to support 128 KB of RAM.
Which analysis correctly evaluates whether this is possible?

○ **A.** Possible: 16 bits can address 16 × 1024 = 16,384 locations, and 128 KB needs only 128 locations.

○ **B.** Not possible: A 16-bit address bus can only address 2^16 = 65,536 bytes (64 KB), which is less than 128 KB.

○ **C.** Possible: The system can use banking or paging to access memory beyond the direct addressing limit.

○ **D.** Not possible: 128 KB requires at least a 32-bit address bus because 128 is close to 256.

Q29 (3 marks)

A simple CPU executes the fetch-decode-execute cycle where:
- Fetch stage: 2 clock cycles
- Decode stage: 1 clock cycle
- Execute stage: 2 clock cycles

How many clock cycles are needed to execute 4 instructions sequentially (without any overlap)?

○ **A.** 5 cycles, because each instruction takes 5 cycles and they share resources

○ **B.** 20 cycles, because each instruction takes 5 cycles (2+1+2) and 4 instructions run sequentially

○ **C.** 8 cycles, because only the execute stage matters for instruction count

○ **D.** 4 cycles, because one cycle completes one instruction

Q30 (3 marks)

A software developer profiles their application and finds:
- 40% of execution time spent on memory access operations
- 35% spent on integer calculations
- 25% spent on floating-point calculations

The developer considers three upgrade options:
Option A: Faster RAM that reduces memory access time by 50%
Option B: CPU with 2× faster integer ALU (halves integer calculation time)
Option C: CPU with 2× faster floating-point unit (halves floating-point calculation time)
Which upgrade provides the greatest performance improvement?

○ **A.** Option A, reducing total execution time by 20% (0.5 × 40% = 20% improvement)

○ **B.** Option B, reducing total execution time by 17.5% (0.5 × 35% = 17.5% improvement)

○ **C.** Option C, reducing total execution time by 12.5% (0.5 × 25% = 12.5% improvement)

○ **D.** All options provide equal improvement because the ratios are similar

Total: **55** marks • Time: **30 min**

↑ Back to top

Submit answers

Williams Physics Education — over **1,100,000+ marks** of exam-aligned content and **4,200+ auto-marked assessments** across 67 qualifications at GCSE, A-Level, AP & Professional Qualifications. Free to use, no login required.

WPE is free for everyone. If our resources help you or your students, you can support the project here.
© 2026 Williams Physics Education. All rights reserved.